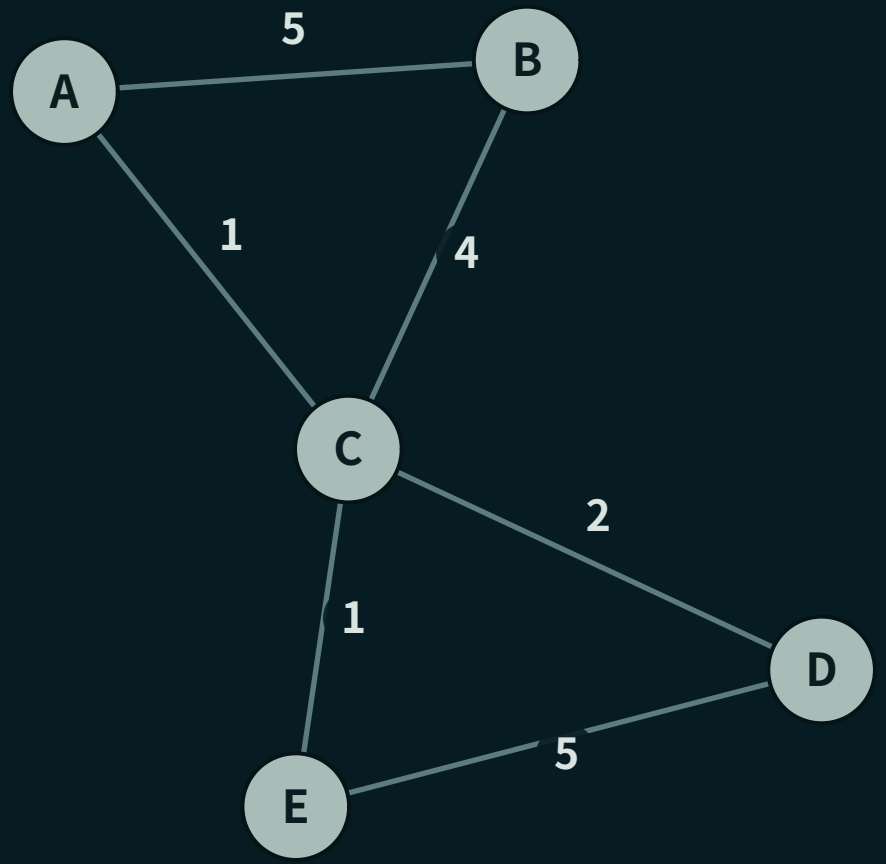
Who is closest?

Affinity metrics · direct weight, shared neighbours and pin-count

Learning Clustering · original author Capeur (Issue #1)

The problem

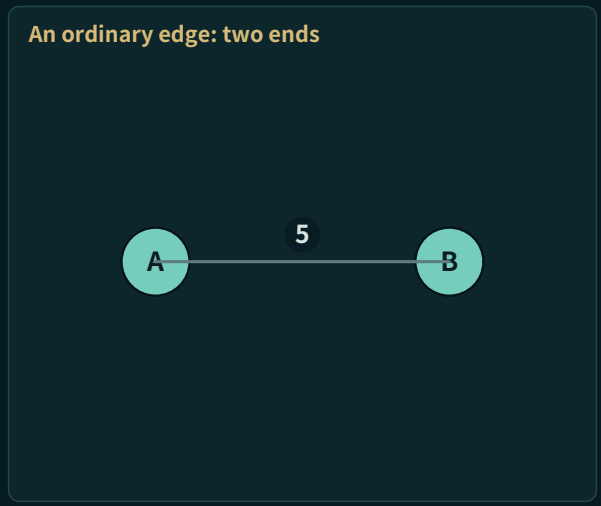
- Input: a weighted undirected graph, or a multi-pin netlist
- Output: every positive pair (u, v, score), with u < v
- Order: score descending, then u and v alphabetically
- Conventions: no self-loops or duplicate edges; a pin repeated in a net counts once; zero pairs are not listed
- A score is a ranking, not a grouping, a distance or a probability

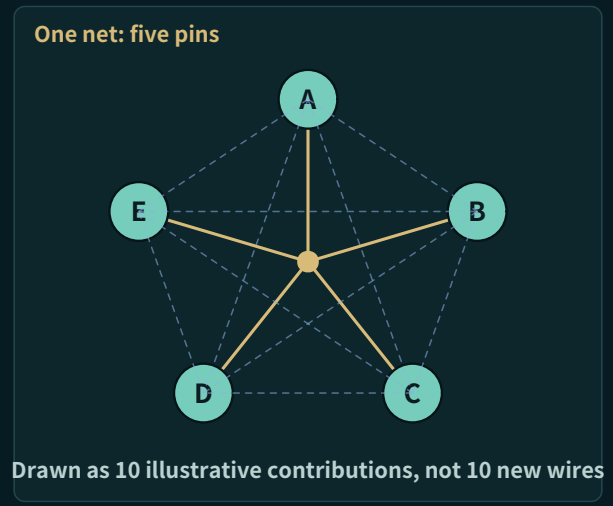


Course graph G01 (total weight 18)

Graphs and nets

- An ordinary edge connects only two cells
- A net can connect the pins of 3, 5 or thousands of cells
- Scoring one pair inside a net needs a model first
- The clique model: draw the net as pairwise illustrative contributions





The pairwise lines are for scoring only; they are not new physical wires.

Direct weight and the stable ranking

- edge(u, v) = w(u, v): only directly wired pairs score
- G01 has 6 edges, so 6 pairs
- A–B and D–E are both 5: by name, A < D
- With the tie rule fixed, the ranking does not depend on input order

Rank	Pair	Score
1	A–B	5
2	D–E	5
3	B–C	4
4	C–D	2
5	A–C	1
6	C–E	1

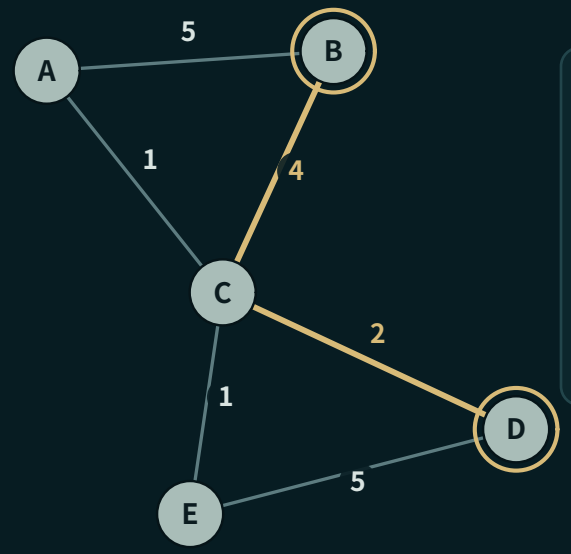
The shared-neighbour score

- shared(u, v) = w(u, v) + Σ_x min(w(u, x), w(v, x))
- x is a shared neighbour, connected to both u and v
- A link through x is only as strong as its weaker edge: take min
- Check every unordered pair: G01 has 10, all positive

Pair	Direct	Via shared	Score
A–B	5	C:1	6
D–E	5	C:1	6
A–C	1	B:4	5
B–C	4	A:1	5
C–D	2	E:1	3
C–E	1	D:2	3
B–D	0	C:2	2
A–D	0	C:1	1
A–E	0	C:1	1
B–E	0	C:1	1

By hand: A–C and B–D

- A–C: direct 1, via B min(5, 4) = 4, total 5
- B–D: no direct edge, so the direct part is 0
- B–D: via C min(4, 2) = 2, total 2
- Score only the existing edges and you miss B–D

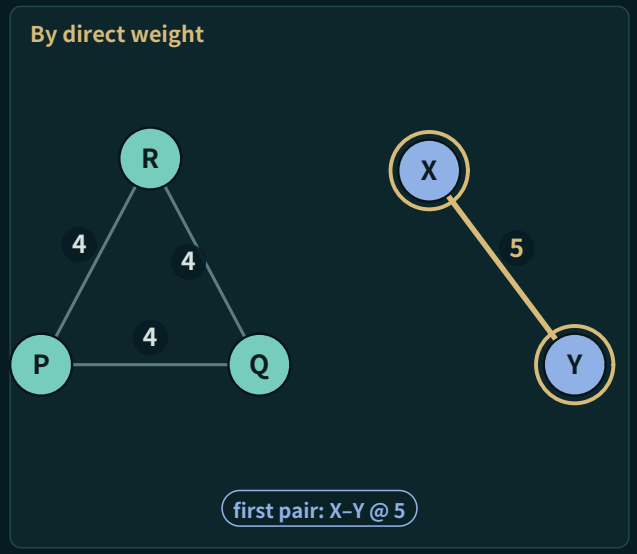


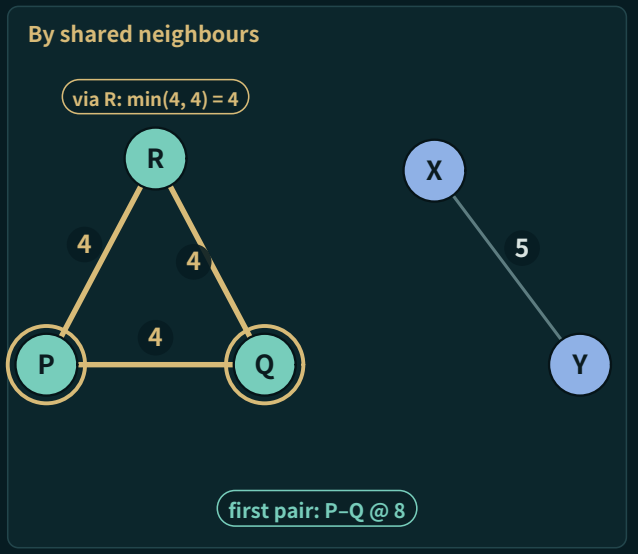
B–D

direct edge w(B,D)	0
via C: min(4, 2)	2
Total	**2**

Change the score and the first pair changes

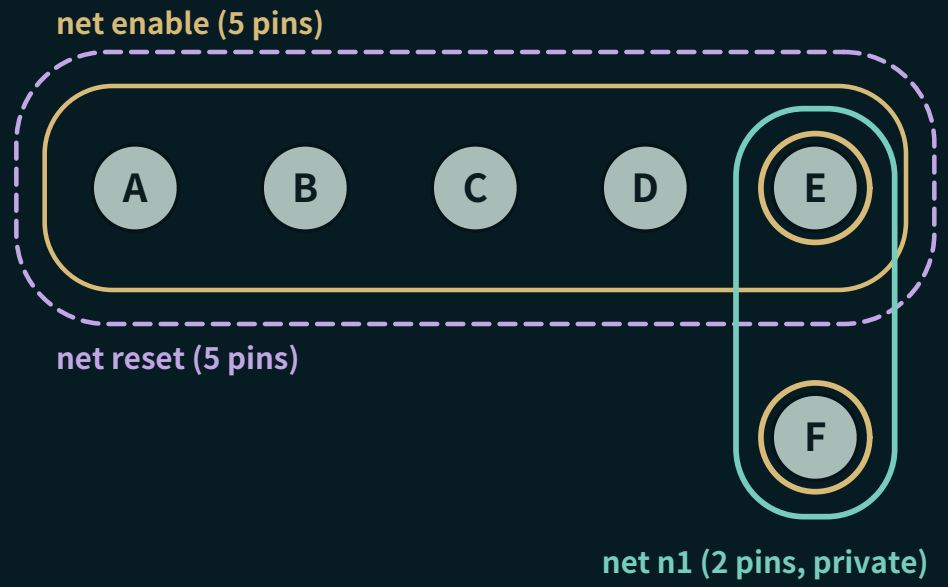
- The priority graph (original author Capeur): triangle PQR with every edge 4, X–Y at 5
- Direct weight: first pair X–Y @ 5
- Shared neighbours: P–Q = 4 + min(4, 4) = 8
- P–Q, P–R and Q–R tie at 8; by name, P–Q
- X–Y needs 9 to be first again (at 8 it loses the tie to P–Q)





How wide nets dilute: pin-count

- Tiny netlist: two five-pin nets connect A–E; the private net n1 connects only E and F
- By net count: A–B = 2, E–F = 1
- pin-count = Σ w(e) / (|e| − 1)
- A–B = 1/4 + 1/4 = 0.5; E–F = 1/(2 − 1) = 1
- Shared out, the private link E–F comes first



	net count	pin-count
A–B	2	0.5
E–F	1	1

The pseudocode, line by line

- Line 1: visit every unordered pair, including pairs with no direct edge
- Line 2: with no edge, start from 0
- Lines 3–4: add the weaker edge for each shared neighbour
- Line 5: record positive scores only (a stated policy)
- Lines 6–7: stable sort; output the ranking

```
1  for each unordered pair (u, v), u < v:
2      score ← w(u, v), or 0 with no edge
3      for each shared neighbour x:
4          score ← score + min(w(u,x), w(v,x))
5      if score > 0: record (u, v, score)
6  sort by (−score, u, v)
7  return the ranking (not a grouping)
```

Common mistakes

- Shared neighbours on existing edges only: missing B–D and 3 other pairs
- Thinking every clustering algorithm merges the top pair first: LP, spectral and KL/FM do not
- Not sharing wide nets out: A–B at 2 beats the private E–F
- Ties that depend on input order: A–B and D–E swap places
- Treating affinity as a distance or a probability

Back to the chip: meaning and limits

- The score decides what gets glued first, which shapes later merging and long cross-region wires
- Say where edge weights come from: wire counts, signal importance, or a conversion from nets
- The small graphs in this course are invented teaching values, not chip measurements
- Shared neighbours check about $n(n-1)/2$ pairs; large netlists must enumerate along adjacency
- The clique model is only one approximation of a multi-pin net

Summary

A score ranks; it does not group. Say which model you use.

Direct weight looks at one wire, shared neighbours at a tight group, pin-count dilutes wide nets.

Next module: the summed weight between clusters as the affinity, and greedy merging begins.